# Sequential convergence of AdaGrad-Norm under local smoothness

**Abstract**

We prove that scalar AdaGrad-Norm converges to a minimizer of every finite-dimensional convex function with full domain, locally Lipschitz gradient, and an attained minimum. Both the learning rate and the positive initial accumulator are arbitrary. The accumulator has a finite limit, and the squared gradient norms are summable. The proof uses a fixed ball around one minimizer: the current-gradient normalization places an auxiliary point in that ball, even before boundedness of the actual iterates is known. A threshold argument and an exact logarithmic product then establish accumulator boundedness. A coupled noncoercive example has super-polynomial gradient growth and satisfies no global generalized-smoothness bound. The result concerns the scalar accumulator; it does not settle the corresponding coordinatewise or full-matrix questions.

**Keywords:** AdaGrad-Norm; local smoothness; convex optimization; sequential convergence; adaptive step sizes

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## 1 The update and the convergence statement

Let $d \geq 1$ be finite and let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and continuously differentiable. Assume that $\nabla f$ is locally Lipschitz and

$$S = \operatorname*{argmin}_{x \in \mathbb{R}^d} f(x) \neq \varnothing. \tag{1}$$

For arbitrary $\eta > 0$, $\delta > 0$ and $x_0 \in \mathbb{R}^d$, define

$$g_k = \nabla f(x_k), \qquad b_{-1} = \delta, \qquad b_k^2 = b_{k-1}^2 + \|g_k\|^2, \tag{2}$$

$$a_k = \frac{\eta}{b_k}, \qquad x_{k+1} = x_k - a_k g_k \qquad (k \geq 0). \tag{3}$$

The current gradient is included in $b_k$ before the update. In particular,

$$a_k \|g_k\| = \frac{\eta \|g_k\|}{\sqrt{\delta^2 + \sum_{i=0}^{k} \|g_i\|^2}} \leq \eta. \tag{4}$$

This is the scalar AdaNorm instance of the template in [2, Algorithm 1].

**Theorem 1.** *Under the stated assumptions, all iterates of* (2)–(3) *are well defined. There exist* $b_\infty < \infty$ *and* $x_\infty \in S$ *such that*

$$b_k \longrightarrow b_\infty, \qquad \sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 = b_\infty^2 - \delta^2 < \infty, \qquad x_k \longrightarrow x_\infty. \tag{5}$$

*There is no upper restriction on* $\eta$ *and no required relation between* $\eta$ *and* $\delta$.

The hypotheses include neither coercivity nor bounded sublevels. No trajectory bound, global Lipschitz constant, Hessian bound, or growth estimate for the gradient is assumed. The conclusion is qualitative: we assert no uniform convergence rate, finite total path length, or specified selection among several minimizers.

### 1.1 Relation to the existing results

Traoré and Pauwels [1] prove sequential convergence for the scalar and coordinatewise AdaGrad variants under global Lipschitz continuity of the gradient. The conclusion of their inspected author version, arXiv:2011.12341v3, conjectures the extension to convex functions with full domain and locally Lipschitz gradient. Theorem 1 establishes the scalar part of that formulation. The comparison to this conjecture is specific to the author version; the distinct typeset journal text has not been inspected here.

Besançon and Le [2, Theorem 5] prove that, for the three AdaGrad variants under convexity and local smoothness, point convergence, boundedness of the trajectory and summability of squared gradients are equivalent. That equivalence does not itself establish any of the three properties for every locally smooth objective. Their Theorem 6 proves scalar convergence under generalized $(L_0, L_1)$-smoothness with $\eta \leq 1/(2L_1)$. Their Theorem 13 allows arbitrary learning rates under the additional polynomially modifiable smoothness condition of their Definition 9. Theorem 1 directly establishes squared-gradient summability for the scalar method throughout the full local-smoothness class. Our final passage from summability to point convergence is the familiar quasi-Fejér argument; the argument below includes it for completeness.

The distinction between the three variants is essential. The proof uses the fact that $a_k$ is a scalar and that the displacement $a_k g_k$ has norm at most $\eta$. It does not supply a result for coordinatewise AdaGrad or full-matrix AdaGrad.

## 2 An inequality from a fixed auxiliary ball

Choose any $z \in S$, and write $f_* = f(z)$. Differentiable convexity gives $\nabla f(z) = 0$. Because $\nabla f$ is locally Lipschitz, it has a finite Lipschitz constant $L \geq 0$ on the compact convex ball

$$B = \{y \in \mathbb{R}^d : \|y - z\| \leq \eta\}. \tag{6}$$

Indeed, a finite cover by local Lipschitz neighborhoods and subdivision of line segments in $B$ give one common constant. This constant concerns the fixed auxiliary ball, not the set of actual iterates.

**Lemma 1.** *For every $k \geq 0$, the actual update satisfies*

$$\langle g_k, x_k - z\rangle \geq f(x_k) - f_* + a_k\|g_k\|^2 - \frac{L}{2}a_k^2\|g_k\|^2, \tag{7}$$

$$\|x_{k+1} - z\|^2 \le \|x_k - z\|^2 - 2a_k\big(f(x_k) - f_*\big) - a_k^2(1 - La_k)\|g_k\|^2. \tag{8}$$

**Proof.** All finite iterates are well defined inductively, since $b_k \ge \delta > 0$ and the gradient is finite everywhere. Define

$$y_k = z + a_k g_k. \tag{9}$$

Equation (4) puts $y_k$ in $B$. The descent lemma on the segment from $z$ to $y_k$, together with $\nabla f(z) = 0$, gives

$$f(y_k) \le f_* + \frac{L}{2} a_k^2 \|g_k\|^2. \tag{10}$$

Convexity at $x_k$, which can lie anywhere in $\mathbb{R}^d$, gives

$$f(y_k) \ge f(x_k) + \langle g_k, y_k - x_k\rangle = f(x_k) + a_k\|g_k\|^2 - \langle g_k, x_k - z\rangle. \tag{11}$$

Combining the last two inequalities proves (7). Expanding

$$\|x_{k+1} - z\|^2 = \|x_k - z\|^2 - 2a_k\langle g_k, x_k - z\rangle + a_k^2\|g_k\|^2 \tag{12}$$

then proves (8). ∎

The coefficient in the final term is exactly $1 - La_k$. It can be negative during early updates. The argument does not claim global objective descent or impose a condition on the initial step; it treats this finite startup through a threshold alternative.

# 3 Bounded accumulation and convergence

## 3.1 The threshold and the logarithmic product

If $L > 0$ and

$$b_k < 2\eta L \qquad \text{for every } k, \tag{13}$$

the nondecreasing sequence $(b_k)$ is already bounded. Otherwise choose a finite $K$ with $b_K \ge 2\eta L$. When $L = 0$, take $K = 0$. In this second case, for all $k \ge K$,

$$La_k \le \frac{1}{2}. \tag{14}$$

Since every objective gap is nonnegative, summing (8) and using (14) yields

$$\frac{\eta^2}{2} \sum_{k=K}^{N} \frac{\|g_k\|^2}{b_k^2} \le \|x_K - z\|^2 \qquad (N \ge K). \tag{15}$$

Set

$$t_k = \frac{\|g_k\|^2}{b_k^2}. \tag{16}$$

The finite prefix and (15) give $\sum_{k\ge 0} t_k < \infty$. Moreover, $0 \le t_k < 1$, because the preceding accumulator is strictly positive. The exact identity

$$1 - t_k = \frac{b_{k-1}^2}{b_k^2} \quad \Longrightarrow \quad \log\frac{b_N^2}{\delta^2} = \sum_{k=0}^{N} -\log(1 - t_k) \tag{17}$$

is more useful here than a one-sided integral estimate. Since $t_k \to 0$, eventually $t_k \le 1/2$ and

$$-\log(1-t_k) \le 2t_k. \tag{18}$$

The finitely many earlier logarithms are finite. Thus the right side of (17) is uniformly bounded, proving boundedness of $(b_N)$ in the second case as well.

In either threshold alternative, monotonicity gives a finite limit $b_\infty$. Telescoping (2) now gives

$$\sum_{k=0}^{\infty} \|g_k\|^2 = b_\infty^2 - \delta^2 < \infty. \tag{19}$$

In particular $g_k \to 0$. No bound on the iterates was used to reach this conclusion.

### 3.2 The limit is one minimizer

For every fixed $z \in S$, convexity gives $\langle g_k, x_k - z\rangle \ge 0$. Therefore (12) and $a_k \le \eta/\delta$ imply

$$\|x_{k+1} - z\|^2 \le \|x_k - z\|^2 + \frac{\eta^2}{\delta^2}\|g_k\|^2. \tag{20}$$

The errors on the right are summable by (19). Adding their remaining tail to the squared distance gives a nonnegative nonincreasing sequence. Hence the distance to each fixed minimizer has a finite limit, and the whole trajectory is bounded.

Finite-dimensional compactness supplies a subsequence $x_{k_j} \to \bar{x}$. By gradient continuity and $g_k \to 0$, one has $\nabla f(\bar{x}) = 0$, so $\bar{x} \in S$. The distance to this fixed minimizer has a limit by (20); along the chosen subsequence that distance tends to zero. It therefore tends to zero along the entire sequence. This proves Theorem 1.

If a finite gradient vanishes, the corresponding point is a minimizer and the remaining sequence is stationary. Constant objectives and the case $L = 0$ on the auxiliary ball are included in the same proof. No Hessian existence was needed for the theorem.

## 4 A coupled example outside global growth bounds

For $(u, v) \in \mathbb{R}^2$, put

$$r = \sqrt{u^2 + v^2}, \qquad q = r - u, \qquad F(u, v) = \exp(q^4) - 1. \tag{21}$$

The function $q$ is convex and nonnegative. The scalar function $s \mapsto e^{s^4} - 1$ is convex and nondecreasing on $[0, \infty)$, so $F$ is convex. Although $q$ is not differentiable at the origin, $q^4$ is $C^2$ there: away from the origin its derivatives of orders $j = 0, 1, 2$ are homogeneous of degree $4 - j$ and bounded on the unit circle. They extend continuously by zero at the origin. Thus $F \in C^2(\mathbb{R}^2)$ and its gradient is locally Lipschitz.

Its minimizer set is the noncompact ray

$$\operatorname{argmin} F = \{(u, 0) : u \ge 0\}. \tag{22}$$

In particular $F$ is not coercive. It is genuinely coupled: for $v > 0$,

$$F_{uv}(0, v) = -(12v^2 + 16v^6)e^{v^4} \neq 0. \tag{23}$$

Along $(u, v) = (-t, 0)$ with $t > 0$, direct differentiation gives

$$F(-t, 0) = e^{16t^4} - 1, \tag{24}$$

$$\|\nabla F(-t, 0)\| = 64t^3 e^{16t^4}, \tag{25}$$

$$F_{uu}(-t, 0) = (192t^2 + 4096t^6)e^{16t^4}. \tag{26}$$

Consequently

$$\frac{F_{uu}(-t, 0)}{\|\nabla F(-t, 0)\|} = \frac{3}{t} + 64t^3 \longrightarrow \infty. \tag{27}$$

There are no fixed finite nonnegative constants $L_0, L_1$ such that

$$\|\nabla^2 F(x)\|_{\mathrm{op}} \leq L_0 + L_1\|\nabla F(x)\| \qquad (x \in \mathbb{R}^2). \tag{28}$$

Indeed, dividing this proposed bound along the displayed ray by the diverging gradient norm contradicts (27). The gradient also exceeds every polynomial bound in the radius. In particular this example does not satisfy the global polynomially modifiable condition of [2, Definition 9], which requires a family of inequalities of the form (28).

Nevertheless, Theorem 1 applies to (21) from every initial point and for every $\eta, \delta > 0$. The example only illustrates the scope of the theorem; it is not used in its proof. The selected point on the ray may depend on the initial point and parameters, and no specific selection formula is asserted.

# References


[1] C. Traoré and E. Pauwels, *Sequential convergence of AdaGrad algorithm for smooth convex optimization*, Operations Research Letters **49** (2021), 452–458. doi:10.1016/j.orl.2021.04.011. Inspected author version: arXiv:2011.12341v3.

[2] M. Besançon and T. Q. Le, *On the iterate convergence of AdaGrad for generalized smooth convex optimization*, arXiv:2608.00482v1 (2026).